Similarity-Cue Injection for Flexible Frame Selection in Video Question Answering

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GitHub: `https://github.com/manishhingar28/DLCV_COURSE_PROJECT`

I. Abstract (Maximum 100 Words)

1. Video-language models are costly because every input frame passes through a large model.
2. So systems feed a few uniform frames, which can miss the key moment or waste budget.
3. Buch et al. (CVPR 2025) place a light Flexible Frame Selector (FFS) before the large model: it scores frames against the question, keeps the useful ones via a differentiable top-k, and uses learnable `[CTRL]` tokens to decide *how many* ($k{-}c$ to k).
4. We read this as learning, from the answer loss alone, which evidence a question needs, so fewer frames match dense sampling.

II. Proposed Novelty & Technical Contribution

1. **Similarity-cue injection** (new module, 0.07M params): FFS learns frame relevance from the answer loss alone through a straight-through top-k, a weak signal with frozen EVA02-CLIP-L/14 features (its top-4 is only +0.7 over uniform). We compute four *zero-shot relevance cues* per frame from the same features (cosine to the question, best and second-best "question + option" similarity, their gap), standardise them per video, and feed them to the frame tokens and, via a linear skip, to the selection logit.
2. Type: architectural modification (new cue module at the selector input); encoder, `[CTRL]`, Gumbel top-k, losses and downstream model unchanged; no labels, no extra network.
3. Difference: FFS learns question–frame similarity through attention from scratch; ours starts from a zero-shot ranker and learns a correction. Both share our FFS recipe (`[CTRL]` count hinge from fixed-k policies).

III. Theoretical / Mathematical Justification

1. $s_{mt} = \langle \hat{t}_m, \hat{f}_t \rangle, \quad c_t = \big[\langle \hat{q}, \hat{f}_t \rangle,\ s_{(1)t},\ s_{(2)t},\ s_{(1)t} - s_{(2)t}\big], \quad \tilde{c}_t = (c_t - \mu_{\text{vid}})/\sigma_{\text{vid}}, \quad l_t = g_\theta\big(h_t + \phi(\tilde{c}_t)\big) + w^\top \tilde{c}_t.$
2. $\hat{f}_t, \hat{q}, \hat{t}_m$: normalised frame, question and "question + option m" features; $s_{(1)}, s_{(2)}$: two best options; h_t: frame token; g_θ: FFS encoder/head; l_t: selection logit. $w{=}0, \phi{=}0$ gives FFS, so our class contains the baseline. **Residual learning:** without cues, relevance must be learned as a bilinear map $\langle W_Q \hat{q}, W_K \hat{f}_t \rangle$ in 768-d from noisy straight-through gradients; the cues give it in closed form ($g_\theta{=}0$ ranks by $w^\top \tilde{c}_t$), so g_θ learns only a correction (lower sample complexity). **Margin = evidence:** with $p(m|f_t) \propto e^{s_{mt}/T}$, a large $s_{(1)t} - s_{(2)t}$ means low entropy $H(a|q, f_t)$: a frame that *discriminates* answers, found without the label. **Standardisation:** top-k depends only on the order of l_t, unchanged by per-video affine shifts, while CLIP cosines shift strongly across videos (mean 0.89); $\tilde{c}_t$ removes this nuisance.
3. Causal link: better ranking ⇒ the same k frames carry more answer evidence ⇒ higher accuracy at equal cost (tested in Table 1). Ablation (RAS): greedy $i_t = \arg\max_i l_i - \beta \sum_{j \in A_{t-1}} S_{ij}$ (submodular), assuming near-duplicate frames waste budget.

IV. Comparative Results & Empirical Evaluation

1. NExT-QA multiple choice, val (4,996 Qs; train 34,132), 32-frame pool, frozen EVA02-CLIP-L/14 features at 224 px; a small frozen QA model (2.0M params) replaces the paper's 8B VLM, so absolute accuracy is not comparable. Shared downstream model and recipe; tuning on held-out *training* videos only; 3 seeds.
2. Reproduced baseline (FFS-style input) vs ours at 4 frames and with adaptive FFS (1–4); Frames = frames given to the downstream model.
3. Takeaway: +1.05 accuracy at 4 frames and +0.85 at 1 frame for +0.07M parameters; with cues, 4 selected frames beat all 32 uniform frames at 1/8 of the downstream cost.

Table 1: Quantitative comparison of the baseline and proposed method on NExT-QA (val, 3 seeds). Best results are highlighted in bold.

Method	Acc (% ↑)	Causal (↑)	Temporal (↑)	Descr. (↑)	Frames (↓)	Params (M)
Uniform sampling, 32 frames / 4 frames	48.40 / 47.94	46.9 / 46.8	46.3 / 45.1	57.9 / 57.7	32 / 4	–
Baseline: Top-4 (FFS-style input)	48.66 ± 0.19	47.8	45.7	57.7	4	2.05
Ours: Top-4 + similarity cues	**49.71 ± 0.10**	**48.3**	**47.4**	**59.1**	4	2.12
Baseline: FFS [paper, re-impl.], adaptive 1–4	46.35 ± 0.19	44.9	44.0	56.2	**1.0**	2.05
Ours: FFS + similarity cues, adaptive	47.20 ± 0.10	45.4	45.0	57.6	**1.0**	2.12
Ablation: Top-4 + cues + RAS	49.71 ± 0.10	48.3	47.3	59.3	4	2.12
Δ (Ours – Baseline), top-4 / FFS	+1.05 / +0.85	+0.5 / +0.6	+1.7 / +1.0	+1.4 / +1.4	same	+0.07

Result Interpretation: Cues improve both selectors in every question type (most: Temporal) by about 5× the seed std at the same cost; with cues, 4 selected frames *beat* all 32 uniform frames (+1.3) at 1/8 of the cost, while the FFS-style input only matches them (+0.3), confirming that cues improve *which* frames are chosen. Trade-offs: adaptive selectors settled on one frame (this downstream model gains only 3.0 points from 1→32 frames); RAS left picks unchanged, and learned picks are *more* similar than uniform ones (0.92 vs 0.89) yet more accurate, so evidence clusters around key moments. Caveats: no per-question significance test; small downstream model; selector reads all 32 frames' features.